

Monoclonal antibody targeting of N-cadherin inhibits prostate cancer growth, metastasis and castration resistance

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Evidence abstract

The transition from androgen-dependent to castration-resistant prostate cancer (CRPC) is a lethal event of uncertain molecular etiology.

Comparing gene expression in isogenic androgen-dependent and CRPC xenografts, we found a reproducible increase in N-cadherin expression, which was also elevated in primary and metastatic tumors of individuals with CRPC.

Ectopic expression of N-cadherin in nonmetastatic, androgen-dependent prostate cancer models caused castration resistance, invasion and metastasis.

Monoclonal antibodies against the ectodomain of N-cadherin reduced proliferation, adhesion and invasion of prostate cancer cells in vitro.

In vivo, these antibodies slowed the growth of multiple established CRPC xenografts, blocked local invasion and metastasis and, at higher doses, led to complete regression.

N-cadherin-specific antibodies markedly delayed the time to emergence of castration resistance, markedly affected tumor histology and angiogenesis, and reduced both AKT serine-threonine kinase activity and serum interleukin-8 (IL-8) secretion.

These data indicate that N-cadherin is a major cause of both prostate cancer metastasis and castration resistance.

Therapeutic targeting of this factor with monoclonal antibodies may have considerable clinical benefit.

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